NOTE: This is the shim needed on the ring gear side for proper backlash.

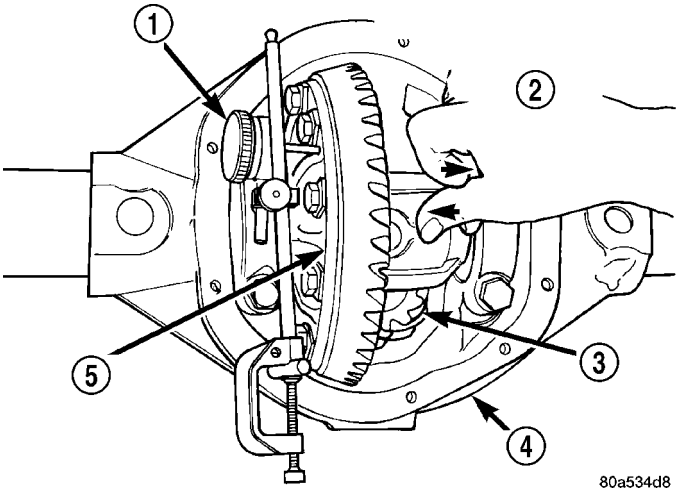


Fig. 22 RECORD DIAL INDICATOR READING

1 - DIAL INDICATOR
2 - DIFFERENTIAL CASE TO RING GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

(21) Subtract the backlash shim thickness from the total preload shim thickness. The remainder is the shim thickness required on the pinion side of the housing.

(22) Rotate dial indicator out of the way on pilot stud.

(23) Remove differential case and dummy bearings from the housing.

(24) Install the selected shims onto the differential case hubs.

(25) Install side bearings on differential case hubs with Install C-3716-A and Handle C-4171.

(26) Install bearing cups on differential.

(27) Install Spreader W-129-B and some items from Adapter Set 6987 on the housing and spread open enough to receive differential case.

CAUTION: Do not spread housing over 0.38 mm (0.015 in.). Failure to heed caution may result in damage.

(28) Install differential case into the housing.

(29) Remove spreader from the housing.

(30) Rotate the differential case several times to seat the side bearings.

(31) Position the indicator plunger against a ring gear tooth (Fig. 23).

(32) Push and hold ring gear upward while not allowing the pinion gear to rotate.

(33) Zero dial indicator face to pointer.

(34) Push and hold ring gear downward while not allowing the pinion gear to rotate. Dial indicator reading should be between 0.12 mm - 0.20 mm (0.005 in. - 0.008 in.). If backlash is not within specifications transfer the necessary amount of shim thickness from one side of the housing to the other (Fig. 24).

(35) Verify differential case and ring gear runout by measuring ring to pinion gear backlash at eight locations around the ring gear. Readings should not vary more than 0.05 mm (0.002 in.). If readings vary more than specified, the ring gear or the differential case is defective.

After the proper backlash is achieved, perform Gear Contact Pattern Analysis procedure.

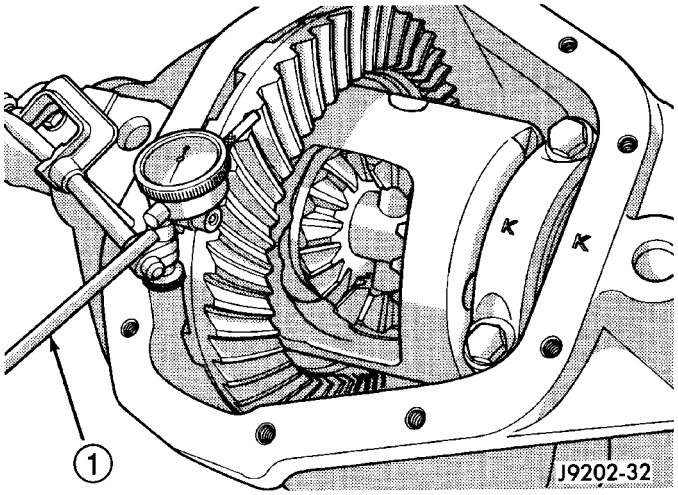


Fig. 23 RING GEAR BACKLASH MEASUREMENT

1 - DIAL INDICATOR

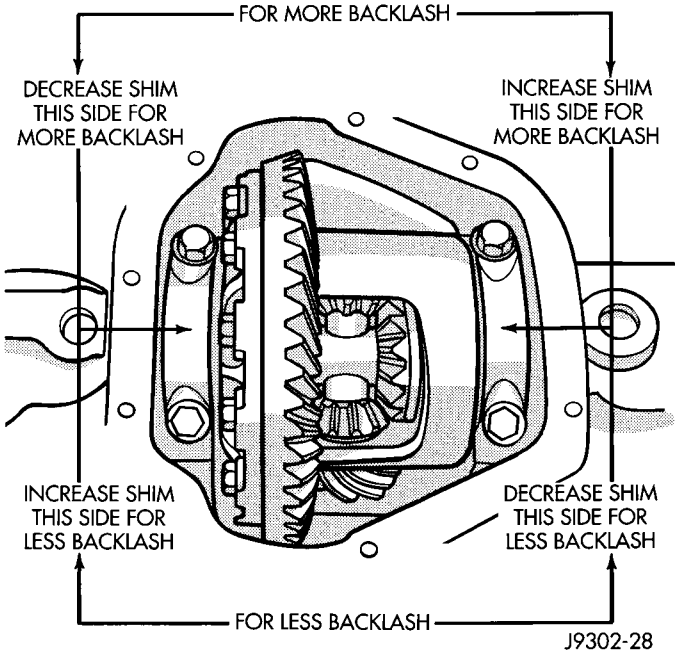


Fig. 24 BACKLASH SHIM

GEAR CONTACT PATTERN

The ring gear and pinion teeth contact patterns will show if the pinion depth is correct in the housing. It will also show if the ring gear backlash has